

DAY 4 – DOCUMENTATION

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1. OBJECTIVE

To understand the differences between Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI Agents, including their architecture, limitations, and role in building applications.

2. RESEARCH TOPICS

2.1. Large Language Models (LLMs)

An LLM is an AI model that understands and generates text.

Examples:

- ChatGPT
- Gemini
- Claude
- Llama
- Mistral

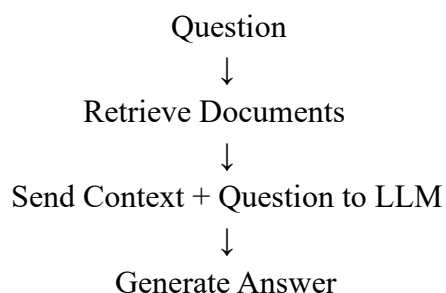
Uses:

- Question answering
 - Content writing
 - Coding assistance
 - Translation
 - Summarization
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2.2. Retrieval-Augmented Generation

RAG retrieves relevant information from external documents and provides it to the LLM before generating an answer.

Workflow:



Uses:

- PDF Chatbot
 - Notes Chatbot
 - FAQ Assistant
 - Knowledge Base Assistant
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2.3. AI Agent

An AI Agent is an LLM-powered system that can reason, use tools, make decisions, and perform actions.

Workflow:**Uses:**

- Research Assistant
 - Coding Assistant
 - Email Assistant
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2.4. Difference Between LLM, RAG, and AI Agent

Feature	LLM	RAG	AI Agent
Meaning	Large Language Model	Retrieval-Augmented Generation	LLM with tools and actions
Main Job	Generate text	Answer using external knowledge	Complete tasks and automate actions
Knowledge Source	Training data	Training data + retrieved documents	Training data + tools + external sources

Feature	LLM	RAG	AI Agent
Uses Documents?	No	Yes	Yes
Uses Tools/APIs?	No	Usually No	Yes
Makes Decisions?	No	No	Yes
Performs Actions?	No	No	Yes
Example	ChatGPT answering a question	PDF Chatbot	Research Assistant

2.5. Limitations

1. Limitations of LLM:

- Can generate incorrect information (hallucination).
- May not know recent information.
- Cannot access private documents by default.
- Same prompt may produce different answers.

2. Limitations of RAG

- Poor chunking can lead to incorrect answers.
- Weak embeddings reduce retrieval accuracy.
- Retrieval may return irrelevant information.
- Retrieved information may still be inaccurate.
- Requires continuous evaluation and monitoring.

3. Limitations of AI Agents

- May choose the wrong tool.
 - Can repeat unnecessary steps or loops.
 - May perform unsafe actions without proper controls.
 - Tool or API failures can affect results.
 - More expensive than simple LLM systems.
 - Requires monitoring and human supervision for important tasks.
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3. Tasks Completed / What I Learned

- Researched the concepts of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and AI Agents.

- Studied the architecture, workflow, and use cases of LLMs, RAG systems, and AI Agents.
 - Compared the capabilities, knowledge sources, and functionalities of LLMs, RAG systems, and AI Agents.
 - Explored the strengths and limitations of each approach and identified suitable use cases for different applications.
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4. Conclusion

This research provided a clear understanding of the differences between LLMs, RAG systems, and AI Agents. LLMs generate responses using their trained knowledge, RAG systems enhance answer quality by retrieving relevant external information, and AI Agents extend these capabilities through reasoning, tool usage, and task execution.